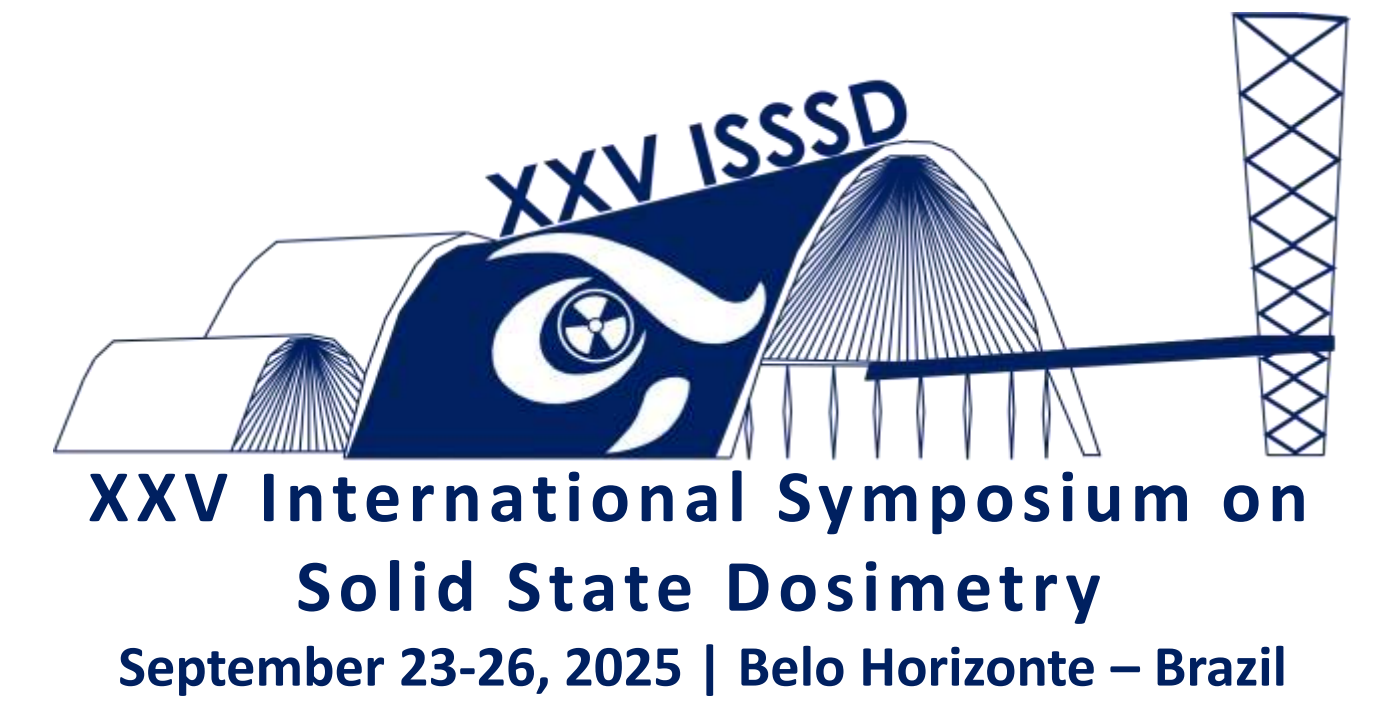


# Adaptation and performance of CdTe-based XFCT detection system using Ag-tacers for Biomedical Applications

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## 1. INTRODUCTION

X-ray fluorescence computed tomography (XFCT) is an imaging technique that enables the mapping of high-atomic-number (Z) elements in biological tissues with high sensitivity and elemental specificity. Although gold (Au) and gadolinium (Gd) nanoparticles have been extensively studied, silver nanoparticles (AgNPs) are emerging as a promising alternative. This is due to their strong K $\alpha$  emission at 22.1 keV, high fluorescence yield (~84%), reduced tissue attenuation, and unique biological effects such as selective cytotoxicity. Its precise detection requires detectors with high spectral resolution in the 20–30 keV range, where cadmium–telluride (CdTe) detectors outperform silicon-based ones. This study evaluated an Amptek XR100 CdTe detector for Ag-XFCT, characterizing it with elemental standards and silver solutions of varying concentrations. Tomographic acquisitions of a PMMA phantom, reconstructed using the MLEM algorithm, successfully demonstrated the feasibility of obtaining 2D silver images for biomedical applications.

## 2. MATERIALS AND METHODS

A conical-beam XFCT system was implemented using a conventional tungsten X-ray tube (60 kV, 5 mA) [1] and an Amptek XR100 CdTe detector [2], positioned at 90° to minimize Compton background. The detector geometry, source-object distance (400 mm), and detector-object distance (40 mm) were fixed, with beam collimation defining the excitation field. Samples were mounted on a motorized platform enabling up to 400 angular steps for 2D acquisitions (Figure 1). Silver nitrate (AgNO<sub>3</sub>) aqueous solutions with concentrations of 0.033%, 0.050%, 0.100% and 5% w/w were prepared in sealed vials to prevent degradation (Figure 2).

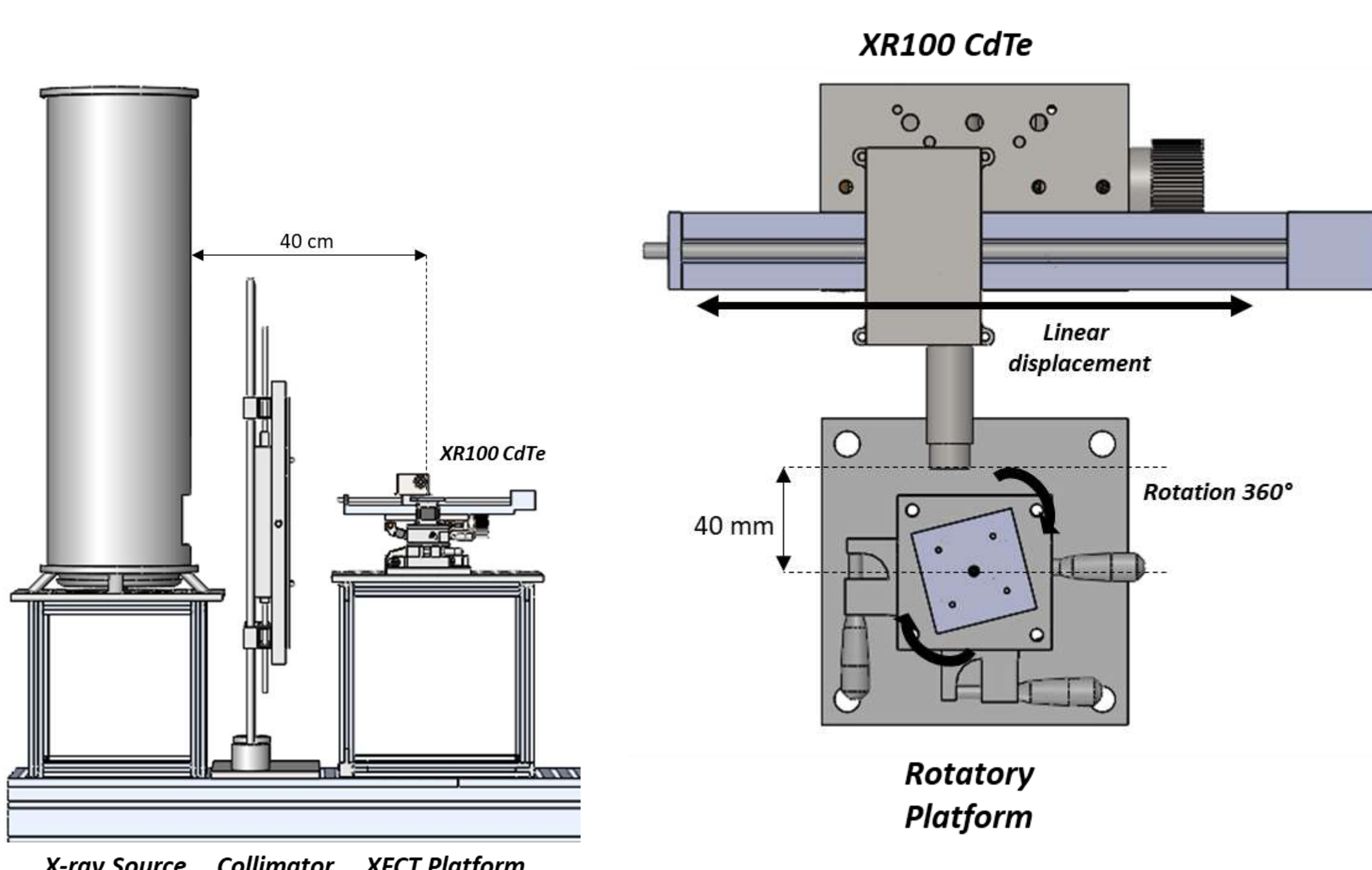


Figure 1.



Figure 2.

The CdTe detector was energy-calibrated using Cu, Pb, W, Gd, and Au fluorescence lines, achieving accurate channel-to-energy conversion. Spectra were acquired (180–300 s) and processed in Python through baseline subtraction, peak fitting, and Gaussian analysis to extract Ag K $\alpha$  and K $\beta$  signals. System sensitivity was evaluated by constructing calibration curves and estimating the limit of detection (LOD) and quantification (LOQ).

$$LOD = \frac{S_{Ag}}{3\sigma} \quad LOQ = \frac{S_{Ag}}{10\sigma} \quad CNR = \frac{S_{AgROI} - S_{bkgROI}}{\sigma_{bkgROI}} \quad (1)$$

where  $\sigma$  is the standard deviation,  $S_{Ag}$  is the Ag signal within the region of interest (ROI) corresponding to the Ag vial, and  $bkg$  is the background signal within the ROI. Spatial resolution was measured by scanning aligned vials at 1 mm steps to obtain fluorescence profiles. Finally, 2D XFCT scans of a PMMA phantom containing Ag samples were performed. Sinograms were reconstructed using the MLEM algorithm (15 iterations), and quantitative image analysis included CNR calculation, applying Rose's criterion (CNR  $\geq$  4) as detectability threshold [3].

## 3. RESULTS

The CdTe-based Amptek detector was calibrated using emission lines from Cu, Gd, Pb, W, and Au, yielding a linear energy–channel relation:  $E \text{ [keV]} = (0.109 \pm 0.001) \times \text{Channel} - (0.49 \pm 0.25)$ , with  $R^2 = 0.9997$  (Figure 3)

Spectral analysis of Ag (5% w/w) revealed well-defined K $\alpha$  (22.16/21.99 keV) and K $\beta$  (24.94 / 25.45 keV) complexes, deconvoluted by multi-Gaussian fitting. Effective resolutions were 0.58 keV (K $\alpha$ ) and 0.75 keV (K $\beta$ ). The measured K $\beta$ /K $\alpha$  intensity ratio (~1:5) agreed with theoretical yields, confirming spectral integrity and linear detector response (Figure 4)

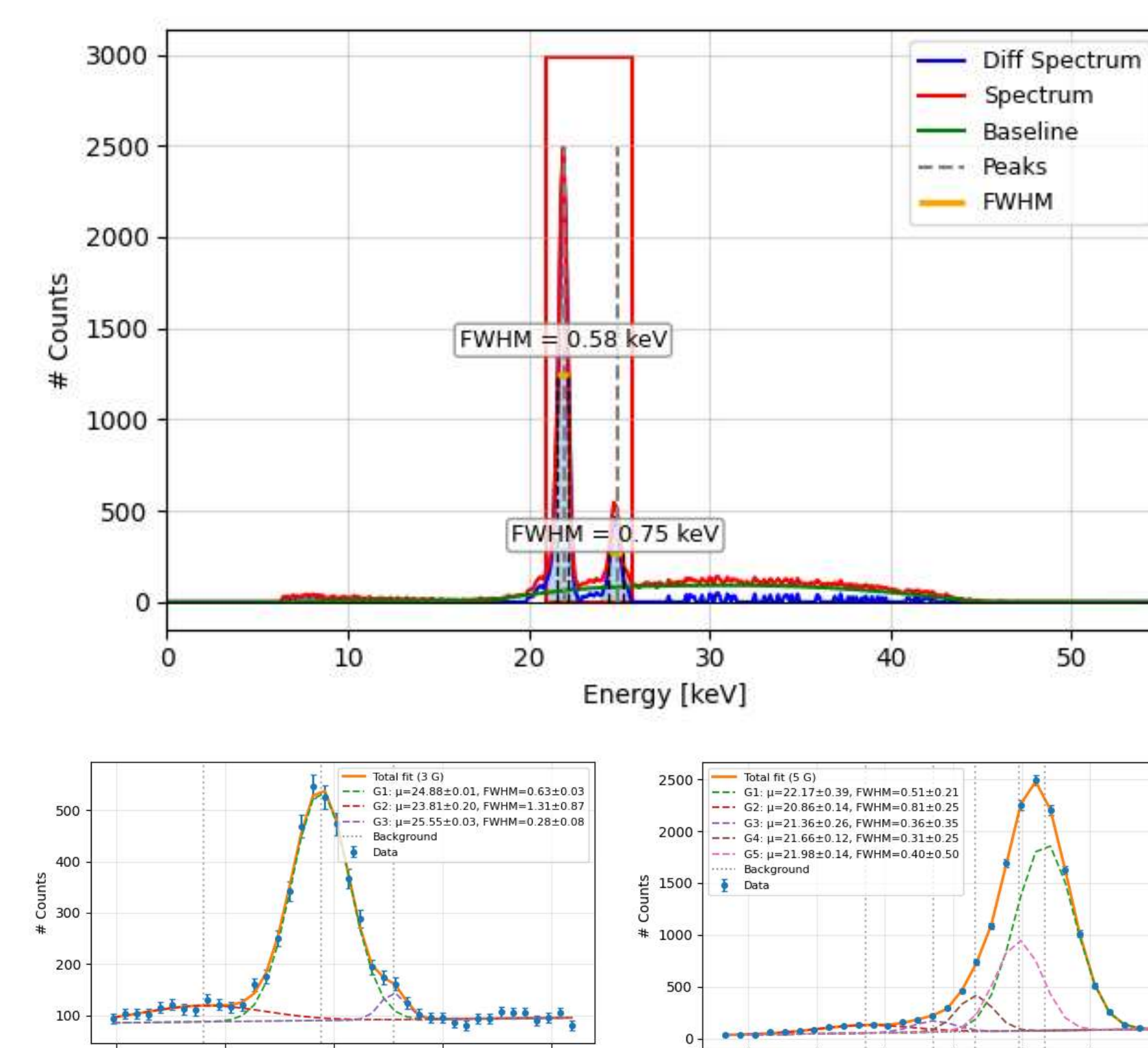


Figure 4.

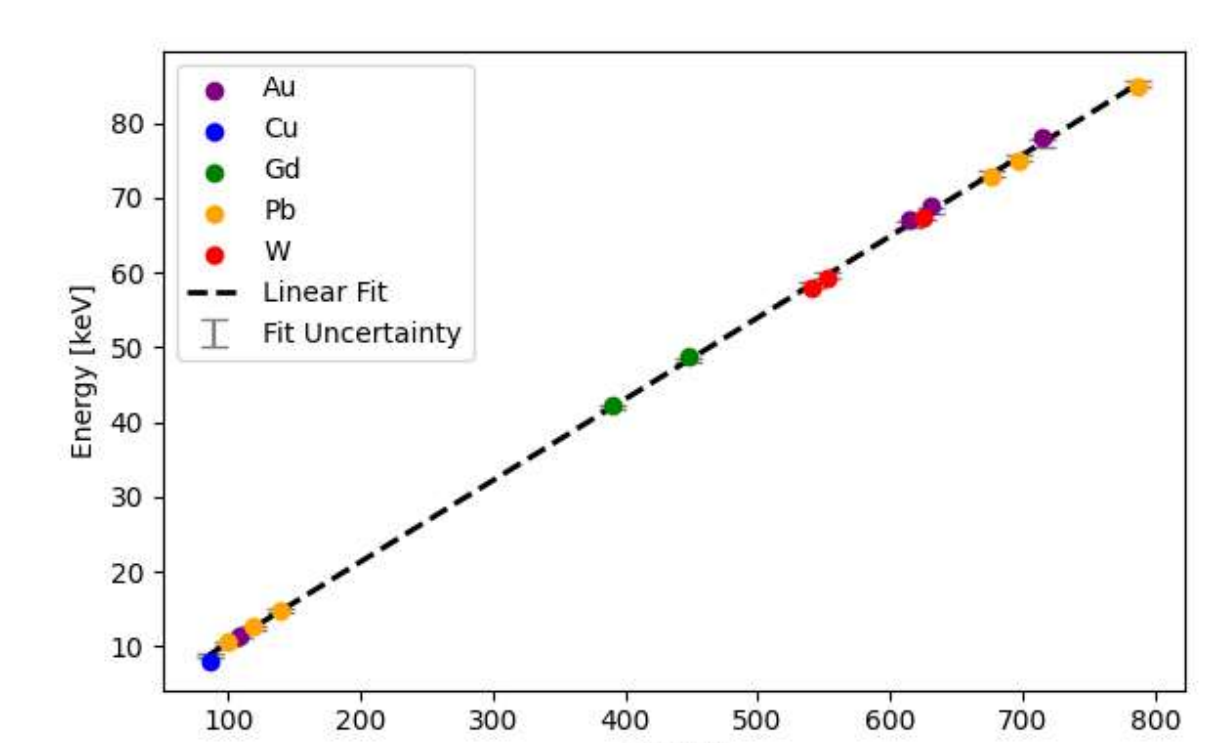


Figure 3.

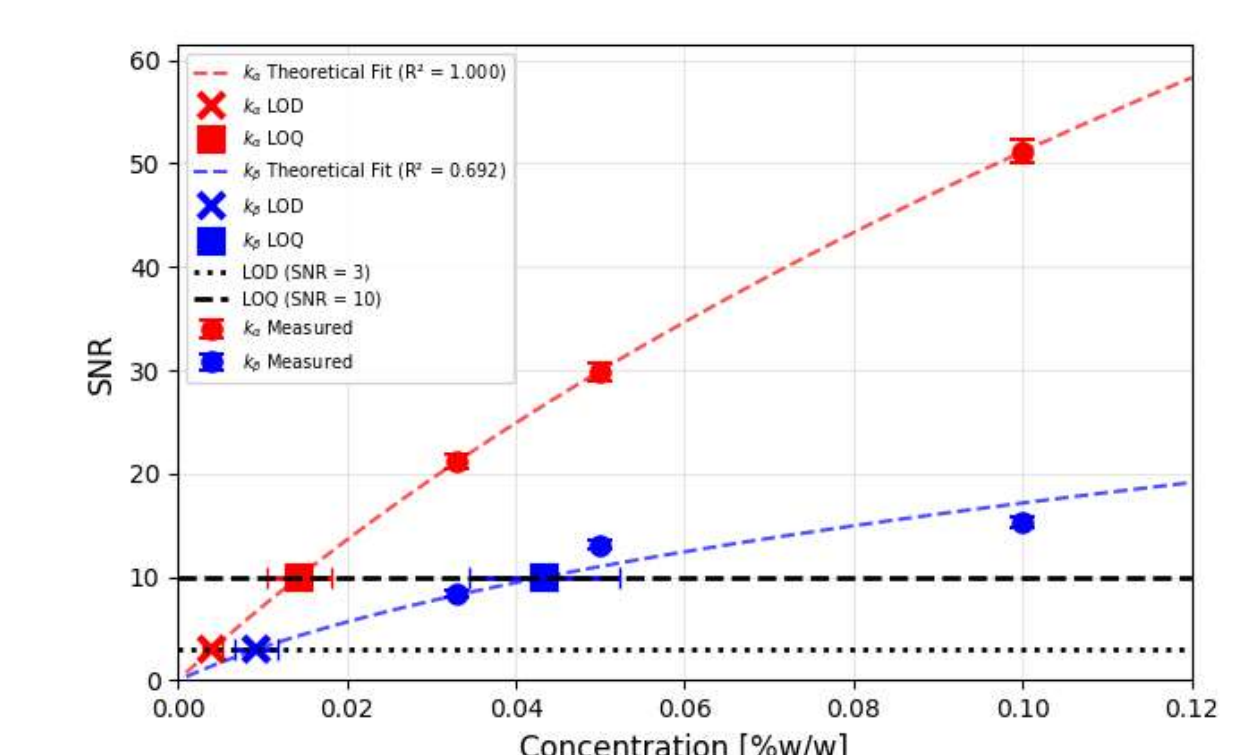


Figure 5.

Sensitivity tests using AgNO<sub>3</sub> solutions showed strong linearity ( $R^2 > 0.999$  for K $\alpha$ ), with LOD of 0.004% w/w and LOQ of 0.015% w/w for K $\alpha$ , outperforming typical AuNP XFCT reports (Figure 5).

A 2D XFCT reconstruction using MLEM (15 iterations) accurately localized vials within  $\leq 0.7$  mm, with linear quantification ( $R^2 = 0.989$ ) (Figure 6)

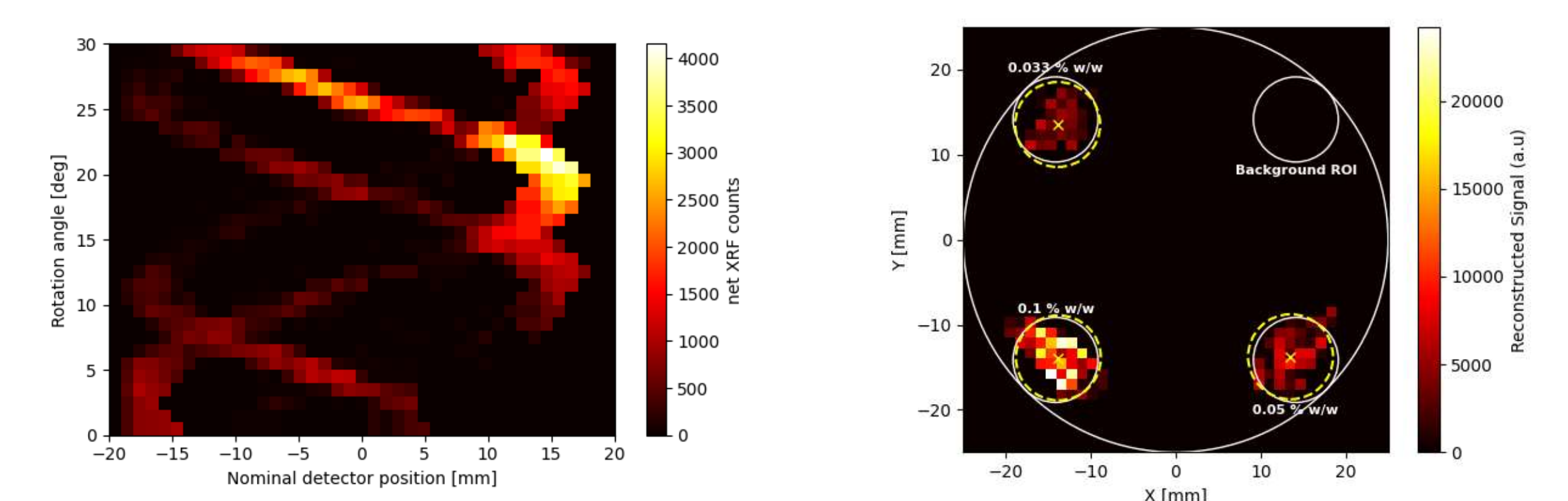


Figure 6.

CNR values of 7.6, 10.7, and 29.5 exceeded Rose's criterion, with a minimum detectable concentration of 0.016% w/w.

## 4. CONCLUSIONS

An X-ray fluorescence computed tomography (XFCT) system was developed and validated for imaging silver agents. The setup, which uses a CdTe detector and iterative reconstruction, achieved high sensitivity (0.016% w/w) and millimeter-scale spatial accuracy. The system, with its simple configuration, proves to be a viable and promising alternative for various studies, pioneering the use of silver tracers.

## References

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